



INDUSTRIAL ENGINEERING & AUTOMATION PRACTICE

Engineering Enquiry Checklist

EXECUTIVE PRACTICE MANDATES & ADVISORY FOCUS

"Essential technical parameters, drawing formats, tolerances, cycle time targets, and factory site constraints required to quote custom machinery accurately."

CORE PRACTICE AREAS INCLUDED IN THIS BROCHURE

■ Component Specifications & Drawings

Technical inputs required to underwrite mechanical concept design.

■ Operational Output & Site Constraints

Production capacity and utility inputs required to size machinery.

■ Preferred Component Makes & Safety Standards

Specifying client-preferred electrical, pneumatic, and control standards.

DIRECT INQUIRY DESK

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01 - Component Specifications & Drawings

Technical inputs required to underwrite mechanical concept design.

Part Geometry & Files

- ☐ **2D Engineering Drawings (PDF / DWG / DXF)**
Showing complete dimensions, critical tolerances, surface finish, and GD&T callouts.
- ☐ **3D CAD Models (STEP / IGES / Parasolid / SolidWorks)**
Native or neutral 3D model files showing complete component geometry.
- ☐ **Raw Material Specification & Hardness**
Material grade (e.g. EN8, SS304, Delrin, ADC12 aluminum) and required heat treatment.
- ☐ **Sample Physical Components (10-20 pieces)**
Mandatory for automated feeding bowl trials, gripper validation, and inspection tests.

Critical Tolerance & Quality Criteria

- ☐ **Identification of High-Precision Features**
Bore concentricity, perpendicularity, surface roughness (Ra value), and flatness limits.
- ☐ **Inspection Protocol & Required CPK**
Target Process Capability Index (e.g. CPK $\geq$ 1.33 or $\geq$ 1.67) on critical dimensions.

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02 - Operational Output & Site Constraints

Production capacity and utility inputs required to size machinery.

Throughput & Process Targets

- ☐ **Target Cycle Time per Part (Seconds / Minutes)**
Maximum allowable floor-to-floor cycle time to meet production volume.
- ☐ **Daily Production Volumes & Shift Pattern**
Planned single-shift (8h), double-shift (16h), or continuous 3-shift operation.
- ☐ **Loading & Unloading Preference**
Manual operator load/unload vs complete bowl-feeder / gantry / robotic automation.
- ☐ **Changeover & Tooling Flexibility**
Whether the machine handles a single part or multiple variant part sizes with rapid changeover.

Factory Site Utilities & Layout Constraints

- ☐ **Available Floor Space & Ceiling Height**
Maximum footprint envelope available on factory floor including maintenance clearance.
- ☐ **Electrical Power Availability**
Available voltage (415V 3-phase / 230V 1-phase), frequency (50Hz), and connected load capacity.
- ☐ **Pneumatic Compressed Air Supply**
Operating pressure (e.g. 6 to 7 bar) and CFM volume capacity at installation point.

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03 - Preferred Component Makes & Safety Standards

Specifying client-preferred electrical, pneumatic, and control standards.

Standard Component Make Preferences

- ☐ **Preferred PLC & HMI Architecture**
Siemens, Allen-Bradley, Mitsubishi, Delta, or Schneider Electric preference.
- ☐ **Preferred Pneumatics & Actuators**
Festo, SMC, or Janatics standardized components.
- ☐ **Electrical Switchgear Standards**
Schneider, Siemens, ABB, or L&T components inside control enclosure.
- ☐ **Machine Safety Compliances**
CE mark compliance, Category 4 safety relays, light curtains, and emergency stop interlocks.

Submit Your Engineering Enquiry

- Email drawings and enquiry checklist to: jyotenterpriseofficial@gmail.com
- Our Senior Mechanical & Automation team will respond with an initial DFM review within 24h
- Direct Engineering Desk: +91 95374 30101 * +91 77780 08999